

Comments on the reviewing report for Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ with no drift term

Le Chen
Auburn University
le.chen@auburn.edu

Nicholas Eisenberg
Auburn University
nze0019@auburn.edu

November 25, 2022

We thank the anonymous referee for careful reading of the paper, some helpful comments, and especially pointing out the reference by Gu and Li [1]. In this revision, we have carefully taken into all comments from the report. The list of major changes are as follows:

1. pages 3 – 4, in the both paragraphs after Eq. (1.9), we have included the reference [1] and made some comparison with our current work. As a short summary, the focus of our paper is on the conditions on the correlation function/measure and initial conditions in order to guarantee the existence of invariant measures. The focus of the paper [1] is different, and convenient conditions were made in [1] for the correlation function, which is assumed to be smooth function with compact support. The initial condition in [1] is essentially constant initial data with some perturbation. In contrast, our conditions on both correlation function/measure and initial conditions are much weaker; see the main theorem – Theorem 1.5 of the paper.
2. page 5, Figure 1 has been simplified and updated. We postponed the conditions formulated using $\mathcal{H}_\alpha(t)$ back in Section 3.1 when it is first used.
3. page 5, Remark 1.2 has been added to comment on the including the *strengthened Dalang's condition* in our main theorem.
4. page 7, Remark 1.6 has been included to comment on our conditions on the noise. Both this change and the previous one are related to one comment from the referee about our additional assumption, namely, the *strengthened Dalang's condition*.
5. Section 3.2, Figure 2 and Remark 3.5 have been created to help the reader. We have also updated Lemma 3.6 accordingly to reflect the changes in items 2, 3, and 4 above.
6. In the Acknowledgment section, we include acknowledgment to Yu Gu for the discussions on the reference [1].

References

- [1] Yu Gu and Jiawei Li. “Fluctuations of a nonlinear stochastic heat equation in dimensions three and higher”. In: *SIAM J. Math. Anal.* 52.6 (2020), pp. 5422–5440. ISSN: 0036-1410.